

Roman Lubkov

Curriculum Vitae

Positions

- 2025–present **Senior Lecturer**, Mathematics and Computer Science Department, St. Petersburg University
- 2025–present **Prime instructor**, St. Petersburg Center for Artificial Intelligence and Information Technologies
- 2023–present **Postdoctoral researcher**, Leonhard Euler International Mathematical Institute, St. Petersburg
- 2022–2023 **Junior researcher**, Chebyshev Laboratory at St. Petersburg University
- 2019–2022 **Research assistant**, St. Petersburg Department of Steklov Mathematical Institute of The Russian Academy of Sciences, St. Petersburg
- 2017–2020 **Research assistant**, Mathematics and Mechanics Department, St. Petersburg University

Summary of Qualifications

- Author of 10+ publications in peer-reviewed mathematical journals, see full list on p. 4;
- Experienced Primary Instructor in advanced mathematics, having taught core undergraduate algebra and specialized PhD seminars for 5+ semesters.
- Awarded multiple research grants as principal investigator and co-investigator, funded by the Russian Science Foundation (RSF) and other agencies;
- Proficient in scientific computing and DevOps, with expertise in Linux server management, containerization, and automated research workflows.

Education

- 2017–2021 **Ph.D. in Mathematics and Physics**, St. Petersburg Department of Steklov Mathematical Institute of The Russian Academy of Sciences
Advisors: Nikolai Vavilov & Victor Petrov *Overgroups of The Elementary Subgroups of Reductive Groups in Irreducible Representations*
- 2015–2017 **M.Sc. in Mathematics**, St. Petersburg University, Diploma Summa Cum Laude
Advisor: Nikolai Vavilov *Overgroups of Exterior Powers of an Elementary Group*
- 2011–2015 **B.Sc. in Mathematics**, St. Petersburg University, Diploma Summa Cum Laude
Advisor: Nikolai Vavilov *Overgroups of Exterior Square of an Elementary Group*

Research Interests

- Algebraic groups: structure theory over rings (Chevalley groups, Steinberg groups, and their relatives; representation theory of such groups)
- Invariant theory: geometry and structure problems defining algebraic groups via invariants; algebro-geometric [Grothendieck] aspects
- Non-associative algebras: Jordan algebras and superalgebras, Poisson algebras and superalgebras and their generalizations, deformations and degenerations

Funding and research groups

- RSF: [Russia] Russian Scientific Foundation
- RFBR: [Russia] Russian Foundation for Basic Research

- 2025–2027 **Co-coordinator**, *New trends in non-associative algebras*, RSF No 25-21-00545, 19.500 US\$, head: Gleb Nenashev, [Link](#)

- 2025 **Participant**, *Analysis, geometry, mathematical physics and their applications*, RSF No 19-71-30002 (extension), head: Anton Baranov, [Link](#)
- 2023–2025 **Leader**, *Groups and non-associative algebras*, RSF No 22-71-10001, 133.000 US\$
Collective research grant (participants: I. Kaygorodov, A. Generalov, A. Petukhov, I. Karimjanov, V. Lopatkin, K. Vyatkina, M. Venchakov), [Link](#)
- 2023–2024 **Participant**, *Non-associative algebraic structures*, RSF No 22-11-00081, head: Ilya Gorshkov, [Link](#)
- 2019–2022 **Co-coordinator**, *Decomposition of unipotents in polyvector representations and applications*, RFBR, individual grant for doctoral students No 19-31-90072, 17.000 US\$, head: Victor Petrov, [Link \(in Russian\)](#)
- 2017–2019 **Participant**, *Split reductive groups over rings and their relatives*, RSF No 17-11-01261, head: Alexei Stepanov, [Link](#)

Awards and scholarships

- 2025 **Researcher Award Competition in Mathematics, Theoretical Computer Science, and Theoretical Physics**, winner, [Link \(in Russian\)](#)
- 2023 **The “School” competition of Theoretical Physics and Mathematics Advancement Foundation “BASIS”**, junior postdoctoral award, [Link \(in Russian\)](#)
- 2022 **Scholarship of the program “Home towns”**, Gazprom Neft PJCS, junior postdoctoral award, [Link \(in Russian\)](#)
- 2018 **Rokhlin Stipends for Young Mathematicians**, graduate student award, [Link \(in Russian\)](#)
- 2013 **Scholarship of Chebyshev Laboratory at St. Petersburg University**, undergraduate student award
- 1999–2011 Multiple winner and prize winner in chess

Visits

- 2025 **Short-term visit of Peking University via Sino–Russian Mathematics Center**, July 29 – August 27, Beijing, China
- **CIMPA school “Modern trends in non-associative algebras”**, April 28 – May 9, Tashkent, Uzbekistan
co-author of poster talk “The algebraic and geometric classification of Jordan superalgebras”
- 2024 **Short-term visit of Peking University via Sino–Russian Mathematics Center**, September 1 – 22, Beijing, China
- 2024 **Mathematics Summer School “Algebraic Transformation Groups III”**, August 5 – 10, Pyalovo, Moscow
lecturer of the mini-course “Why do we need geometry of schemes?”
- 2015 **Scuola Matematica Interuniversitaria (Mathematics Summer School)**, August, Perugia, Italy
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| | Jürgen Appell & | <i>Analisi Funzionale</i> |
| Completed courses: | Martin Văth | |
| | Nikolai Vavilov | <i>Algebra</i> |
- 2015 **Short-term visit of Bielefeld University via Leonhard Euler DAAD program**, April, Bielefeld, Germany, research advisor: Anthony Bak

Conference Talks

- 2024 **Clear about the benefits of schemes for algebraic groups**, *Lie algebras, algebraic groups, and invariant theory*, August 19–24, Samara University, Samara, Russia
- 2023 **Invariants of polyvector representations of $GL_n(\mathbb{R})$** , *Lie algebras, algebraic groups, and invariant theory*, January 28 – February 2, National Research University Higher School of Economics and Steklov Mathematical Institute, Moscow

- 2022 **Plücker polynomials and their applications (poster)**, *Random Point Processes*, November 1–5, Suzdal, Russia
- 2020 **Variation of the inverse decomposition of unipotents for exterior square**, *Lie algebras, algebraic groups, and invariant theory*, January 27 – February 1, Moscow State University and Steklov Mathematical Institute, Moscow
- 2018 **Overgroups of exterior powers of elementary groups (poster)**, *Modern Algebraic Geometry*, July 21–25, Silkroad Mathematics Center, Beijing
- 2017 **Overgroups of exterior powers of an elementary group**, *St. Petersburg Algebraic D. K. Faddeev Seminar*, May 8, St. Petersburg Department of Steklov Mathematical Institute
 - **Overgroups of exterior squares of an elementary group**, *Lie algebras, algebraic groups, and invariant theory*, January 31 — February 5, Moscow State University and Independent University of Moscow

Teaching Experience

- MM SPbU: [Russia] Mathematics and Mechanics Department, St. Petersburg University
 MCS SPbU: [Russia] Mathematics and Computer Science Department, St. Petersburg University
- since 2023 **Abstract algebra**, *undergraduate course, primary instructor*, MCS SPbU, 5 full semesters
 - 2024 Fall **Algebraic groups over rings**, MCS SPbU, organizer of a seminar for PhD students, [Link \(in Russian\)](#)
 - 2024–2025 **Lie algebras**, MCS SPbU, organizer of a seminar for undergraduate students, [Link \(in Russian\)](#)
 - 2014 and 2016, Fall **Abstract algebra**, *undergraduate course, primary instructor*, MM SPbU

Academic Service & Contributions

- MM SPbU: [Russia] Mathematics and Mechanics Department, St. Petersburg University
 MCS SPbU: [Russia] Mathematics and Computer Science Department, St. Petersburg University
- since 2025 **Faculty IT Contributor**, MCS SPbU
 Server administration and infrastructure maintenance for departmental needs; development and management of the official faculty website using a CMS and HTML/CSS.
 - 2025 Fall **Supervising Instructor for a graduate teaching assistant**, MCS SPbU
 - 2015–2017 **Headman of group (25–35 master students)**, MM SPbU
 - 2014 **Coordinator of group (15–25 students)**, MM SPbU
 - 2011–2015 **Headman of group (50+ bachelor students)**, MM SPbU

Other scientific activities and professional development

- since 2026 **Journal referee** for the journal “Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales. Serie A. Matemáticas”.
- since 2026 **Journal referee** for “St. Petersburg Mathematical Journal”.
- since 2025 **Journal referee** for “Communications in Mathematics”.
- since 2024 **Referee** for Theoretical Physics and Mathematics Advancement Foundation “BASIS”, Russia
- since 2023 **Referee** for Student Research Paper Competition, National Research University Higher School of Economics, Moscow

Technical Proficiencies

Infrastructure & DevOps Engineering

Expertise in end-to-end management of Linux servers and self-hosted infrastructure. Experienced in deploying containerized tools (Docker) and automating research workflows.

Scientific Computing & Computer Algebra Systems

Advanced user of Wolfram Mathematica and Maple for symbolic and numerical computations in research

Infrastructure Security & Operations

Practical experience in implementing secure service configurations, VPN tunnels, and maintaining high-availability systems. Working knowledge of core network protocols and security practices for self-hosted environments.

Programming & Typesetting

Familiar with C++ fundamentals. Proficient in $\text{\LaTeX}$ and Typst for professional document preparation.

Additional Competencies

- **Databases:** Operational knowledge of containerized DBMS (MariaDB, MongoDB, Redis).
- **System Administration Skills:** Proficient in managing multi-user Linux environments and configuring complex user workstations.
- **Emerging Tech:** Practical experience with local LLM deployment (Ollama, OpenWebUI).

Publications & Preprints

- [1] **R. Lubkov** and I. Shiringovskiy, "Transposed Poisson structure on the Witt-type algebra $\mathcal{W}(\alpha, -1)$: Derivations, Automorphisms, and Rota–Baxter operators (preprint)," 2026.
- [2] **R. Lubkov**, A. Mordosevich, and A. Stepanov, "Subgroups of general linear groups containing the exterior square of elementary groups (preprint)," 2026.
- [3] H. Abdelwahab, I. Kaygorodov, and **R. Lubkov**, "The algebraic and geometric classification of derived Jordan and bicommutative algebras," *J. Pure Appl. Algebr.*, vol. 230, no. 5, 2026, Art. no. 108252. DOI: [10.1016/j.jpaa.2026.108252](https://doi.org/10.1016/j.jpaa.2026.108252), (Q1, SJR 0.947, WoS, Scopus, RSCI).
- [4] H. Abdelwahab, I. Kaygorodov, and **R. Lubkov**, "The algebraic and geometric classification of right alternative and semi-alternative algebras," *J. Algebr.*, vol. 687, no. 1, pp. 792–824, 2026. DOI: [10.1016/j.jalgebra.2025.09.019](https://doi.org/10.1016/j.jalgebra.2025.09.019), (Q1, SJR 1.029, WoS, Scopus, RSCI).
- [5] H. Abdelwahab, I. Kaygorodov, and **R. Lubkov**, "The algebraic and geometric classification of δ -Novikov algebras," *Boletín la Soc. Matemática Mex.*, vol. 31, no. 3, 2025, Art. no. 145. DOI: [10.1007/s40590-025-00827-4](https://doi.org/10.1007/s40590-025-00827-4), (SJR 0.414, WoS, Scopus, RSCI).
- [6] K. Abdurasulov, **R. Lubkov**, and A. Saydaliyev, "The algebraic and geometric classification of Jordan superalgebras," *Sib. Electron. Math. Reports*, vol. 22, no. 1, pp. 813–852, 2025. DOI: [10.33048/semi.2025.22.052](https://doi.org/10.33048/semi.2025.22.052), (SJR 0.313, Scopus, RSCI).
- [7] **R. Lubkov**, "Reverse Decomposition of Unipotents in Polyvector Representations," *J. Math. Sci.*, vol. 288, no. 3, pp. 350–361, 2025. DOI: [10.1007/s10958-025-07703-6](https://doi.org/10.1007/s10958-025-07703-6), (SJR 0.28, Scopus, RSCI).
- [8] **R. A. Lubkov**, "Overgroups of elementary groups in polyvector representations," Russian, *J. Math. Sci.*, vol. 531, pp. 101–116, 2024, (SJR 0.28, Scopus, RSCI).
- [9] **R. Lubkov** and I. Nekrasov, "Overgroups of exterior powers of an elementary group. Normalizers," *Doc. Math.*, vol. 29, no. 5, pp. 1243–1268, 2024. DOI: [10.4171/dm/956](https://doi.org/10.4171/dm/956), (Q1, SJR 1.035, WoS, Scopus, RSCI).
- [10] **R. Lubkov** and I. Nekrasov, "Overgroups of exterior powers of an elementary group. Levels," *Linear Multilinear Algebr.*, vol. 72, no. 4, pp. 563–584, 2022. DOI: [10.1080/03081087.2022.2160422](https://doi.org/10.1080/03081087.2022.2160422), (Q1, SJR 0.837, WoS, Scopus, RSCI).
- [11] **R. Lubkov**, "The reverse decomposition of unipotents for bivectors," *Commun. Algebr.*, vol. 49, no. 10, pp. 4546–4556, 2021. DOI: [10.1080/00927872.2021.1924183](https://doi.org/10.1080/00927872.2021.1924183), (SJR 0.656, WoS, Scopus, RSCI).
- [12] **R. Lubkov** and A. Stepanov, "Subgroups of Chevalley Groups Over Rings," *J. Math. Sci.*, vol. 252, no. 6, pp. 829–840, 2021. DOI: [10.1007/s10958-021-05203-x](https://doi.org/10.1007/s10958-021-05203-x), (SJR 0.28, Scopus, RSCI).
- [13] **R. A. Lubkov** and I. I. Nekrasov, "Explicit Equations for Exterior Square of the General Linear Group," *J. Math. Sci.*, vol. 243, no. 4, pp. 583–594, 2019. DOI: [10.1007/s10958-019-04559-5](https://doi.org/10.1007/s10958-019-04559-5), (SJR 0.28, Scopus, RSCI).